

Daniel Sinambela

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Employment

Fall-2025-
2022-2025 Assistant Professor of Mathematics, University of Utah, Asia Campus, Incheon, South Korea.
Postdoctoral Associate at New York University Abu Dhabi, UAE.
Mentor: Dr. Weiren Zhao.

Education

2022 Ph.D in Applied Mathematics, University of Missouri, Columbia, USA.
Advisor: Dr. Samuel Walsh.
2018 M.A. (MAGNA CUM LAUDE) in Mathematics, University of Missouri, Columbia, USA.
Advisor: Dr. Samuel Walsh.
2015 B.S. (MAGNA CUM LAUDE) in Mathematics and Minor in Statistics, University of Missouri, Columbia, USA.

Research interest

Main: Fluid Dynamics • Hydrodynamics Stability • Water Waves • Nonlinear PDEs • Harmonic Analysis • Mathematics Education

Publications and Preprints

1. *Small-Amplitude Solitary Waves for f -Plane Capillary-Gravity Flows with Arbitrary Vorticity*, (To Appear on arXiv)
2. *Generalized Error Bounds in the Recovery of Solitary Wave Profiles*, [Link](#).
3. *Internal Capillary-Gravity Solitary Waves in Two-Layer Rotational Fluids with Finite Depth*, (with T Hsiao, K Chen) (submitted).
4. *Periodic interfacial capillary-gravity water waves with arbitrary vorticity* (with L.J Wang, (To appear) Journal Mathematics of Fluid Mechanics).
5. *Asymptotic Stability of the Three-Dimensional Couette Flow for the Stokes-transport Equation* (with W Zhao, R Z Zhu, (To Appear) Archive for Rational Mechanics and Analysis), [Link](#).
6. *On the Asymptotics of Solitary Waves with Compactly Supported Vorticity*, Journal of Differential Equations, [Link](#).

7. *Asymptotic Stability of the Two-Dimensional Couette Flow for the Stokes-transport Equation* (with W Zhao, R Z Zhu, Transactions of the American Mathematical Society), [Link](#).
8. *The Transition to Instability for Stable Shear Flows in Inviscid Fluids* (with W Zhao, Journal of Functional Analysis), [Link](#).
9. *Existence and Stability of Interfacial Capillary-Gravity Solitary Waves with Constant Vorticity*, Indiana University Math J, [Link](#).
10. *Large-Amplitude Solitary Waves in Two-Layer Density Stratified Water*, SIAM J. Math. Anal. 53-4(2021), [Link](#).

Other Publications

1. *Global Bifurcation and Stability of Solitary Waves in Two-Layer Water*. [Link](#)

In preparation

1. *Internal Waves in Axisymmetric Electrohydrodynamic fluids* (with L.J Wang) (in preparation).
2. *Error Analysis in Symbolic and Graphical Understanding of Function Composition* (with S Pathak) (in preparation).
3. *Instability of the Two-Dimensional Couette flow for the Stokes-transport Equation*, (with W Zhao, R Zi) (in preparation).

Teaching Experience

Spring-2026	Instructor for Math 1030 (Quantitative Reasoning) , 1050 (College Algebra) , 1105 (Business Calculus) , University of Utah, Asia Campus, Incheon.
Fall-2025	Instructor for Math 1030 (Quantitative Reasoning) , 1050 (College Algebra) , 1090 (Business Mathematics) , University of Utah, Asia Campus, Incheon.
Fall-2024	Informal Instructor (Substituting) for Math-UH 1000 , New York University, Abu Dhabi.
Fall-2021	Instructor for Math 4100/7100 Ordinary Differential Equations (Online) , University of Missouri, Columbia.
Summer	Instructor for Math 4100/7100 Ordinary Differential Equations , University of Missouri, Columbia.
Spring-2021	Instructor for Math 1700 Calculus II , University of Missouri, Columbia.
Fall-2020	Instructor for Math 1700 Calculus II , University of Missouri, Columbia.
Summer	Instructor for Math 4100/7100 Ordinary Differential Equations , University of Missouri, Columbia.
2019-2020	Instructor for Math 1400 Calculus for Business and Social Science , University of Missouri, Columbia.
Spring-2019	Instructor for two sections of Math 1100 Finite Mathematics , University of Missouri, Columbia.
Fall-2018	Instructor for two sections of Math 1300 College Algebra , University of Missouri, Columbia.
Spring-2018	Teaching assistant for Math 4140 Linear Algebra (Matrix Theory) , University of Missouri, Columbia.
Spring-2018	Teaching assistant for Math 1160 Precalculus , University of Missouri, Columbia.
Fall-2017	Teaching assistant for Math 1500 Calculus I , University of Missouri, Columbia.

Tutoring and Mentoring

- 2024 Helping instructor teach precalculus recitation at NYU Abu Dhabi.
- 2022-2023 Online Math Teacher for Negeri Anak Indonesia: mentoring and tutoring junior and high School students at Solandres orphanage.
- 2019-2020 Math Tutor for Grade A-Plus: tutoring underrepresented group (African-American students) in Columbia, Missouri.
- 2019 Math Tutor at the Math@Case: tutoring college students at University of Missouri, Columbia.
- 2018 Graduate Student Teaching Mentor: mentoring new graduate students in teaching at University of Missouri, Columbia.

Honors & awards

- 2022 Recipient of the **Blumenthal Award** for outstanding dissertation and best academic and research performance.
Recipient of the departmental outstanding **Graduate Teaching Award**.
Recipient of **Travel Award** to attend Workshop and Summer School on New Directions in Water Waves, University of Bath, UK.
- 2019 **Travel Award** to attend Summer School on Recent Advances in Mathematical Fluid Dynamics, USC, California.
- 2018 Graduate students **Mentoring Award**.
- 2015 Recipient of the **Curator Grant Scholarship** to support graduate education given by International Center.
Recipient of the **Promus Fund Scholarship** given by the Department of Mathematics.
- 2014 Finished **4th position in the Missouri Mathematics Collegiate Competition**.
- 2013 Recipient of **Phyllis Ann Heyssel Scholarship** given by the Department of Mathematics.
- 2012 Recipient of **Putera Sampoerna Foundation** Scholarship to study in USA for four years.

Book

- 2021 Math Textbook for Grade 11 High School (joint work with the Indonesian Ministry of Education) [Link](#).

Talks and Posters

- 2024 American Institute of Mathematical Sciences Conference, New York University Abu Dhabi, UAE.
Talk: *The transition to instability for stable shear flows in inviscid fluids*
- 2024 American Institute of Mathematical Sciences Conference, New York University Abu Dhabi, UAE.
Talk: *Existence and stability of interfacial capillary-gravity solitary waves with constant vorticity*
- 2024 Postdoctoral Seminar, New York University Abu Dhabi, UAE.
Talk: *The transition to instability for stable shear flows in inviscid fluids*
- 2022 The 12th IMACS International Conference, University of Georgia, Athens.

- 2021 Talk: *Large-amplitude solitary waves in two-layer density stratified water*
Differential Equations Seminar, University of Missouri, Columbia.
- 2021 Talk: *Large-amplitude solitary waves in two-layer density stratified water*
KUMUNU-ISU Conference on PDE, Dynamical Systems, and Applications, University of Nebraska, Lincoln.
- Poster: *Large-amplitude solitary waves in two-layer density stratified water*

Workshops & Conferences

- 2022 New Directions in water waves, workshop and summer school, University of Bath, England.
- 2022 The 12th IMACS International Conference on Nonlinear Waves at University of Georgia, Athens.
- 2021 KUMUNU-ISU Conference at University of Nebraska, Lincoln.
- 2020 MSRI 2020 on Introduction to Water Waves at UC Berkeley.
- 2019 Summer School on Recent Advances in Mathematical Fluid Dynamics at University of Southern California.
- 2019 KUMUNU Conference at University of Missouri, Columbia.
- 2018 KUMUNU Conference at University of Kansas, Lawrence.

Research Visit

- 2024 Visiting Prof. Ruizhao Zi at Central China Normal University for research collaboration.

Research assistantships

- Summer-2020 Research assistantship , University of Missouri, Columbia,
Supported by National Science Foundation through the award NSF DMS-1812436 (PI: Samuel Walsh).
- Summer-2019 Research assistantship , University of Missouri, Columbia,
Supported by National Science Foundation through the award NSF DMS-1710989 (PI: Yuri Latushkin).

Peer Review Service

PDEs on fluids Journal: Indiana University Mathematics Journal, Annals of Physics.
Mathematics Education Journal: Frontiers in Education.

Professional Memberships

American Mathematical Society (AMS), Society for Industrial and Applied Mathematics (SIAM).

Certification

- 2024 Post Graduate certificate in Data Science and Business Analytics, University of Texas, Austin. [Link](#)
Topics: Statistical Analysis using Python, Exploratory Data Analysis, Data Visualization, Super-

vised and Unsupervised Learning, Ensemble Techniques, Model Tuning.

Computer Skills

PROGRAMMING LANGUAGE:

Python, SAS, R, Matlab.